VIDYAPEETH

BATCH CODE: 12- AN181MA

SUBJECT NAME: Zoology

CHAPTER NAME:
Structural organization in animal.

Lecture No. 06

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Today's Goal

Subtopic .. Types of S. Epithelium

Subtopic .. Gland.

Subtopic ..

Class = Energy

Simple Columnar Epithelium

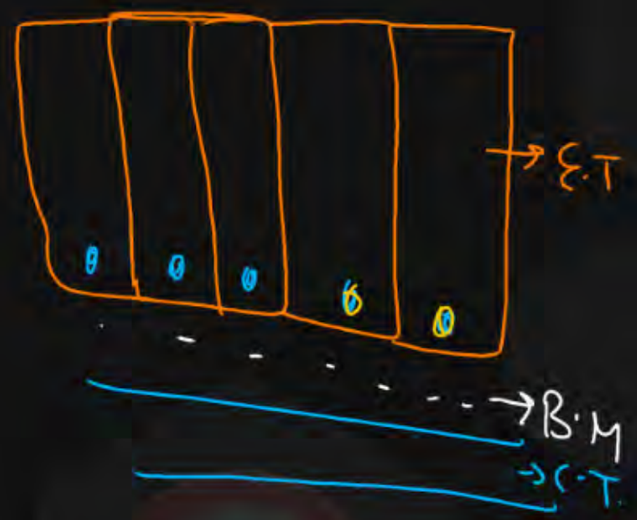
① Shape of cell : Long, Tall, Slender shape

② Shape of Nucleus :- Elongated.

③ Position of Nucleus. = At base

④ Function = Absorption & Secretion.

⑤ Found in.

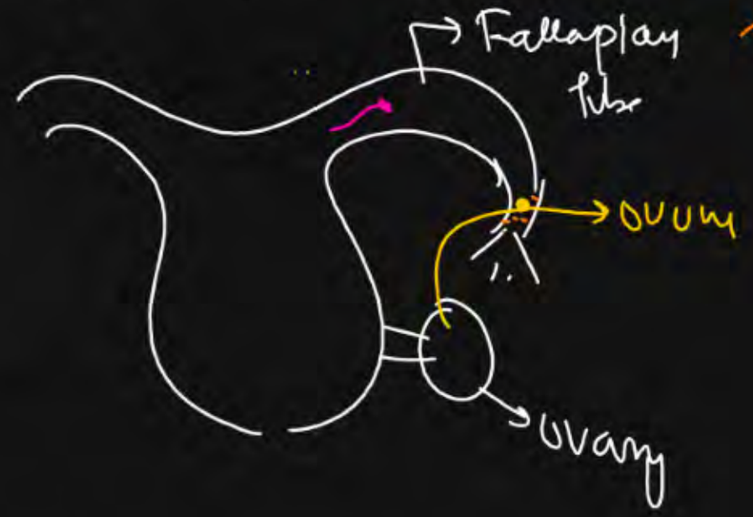


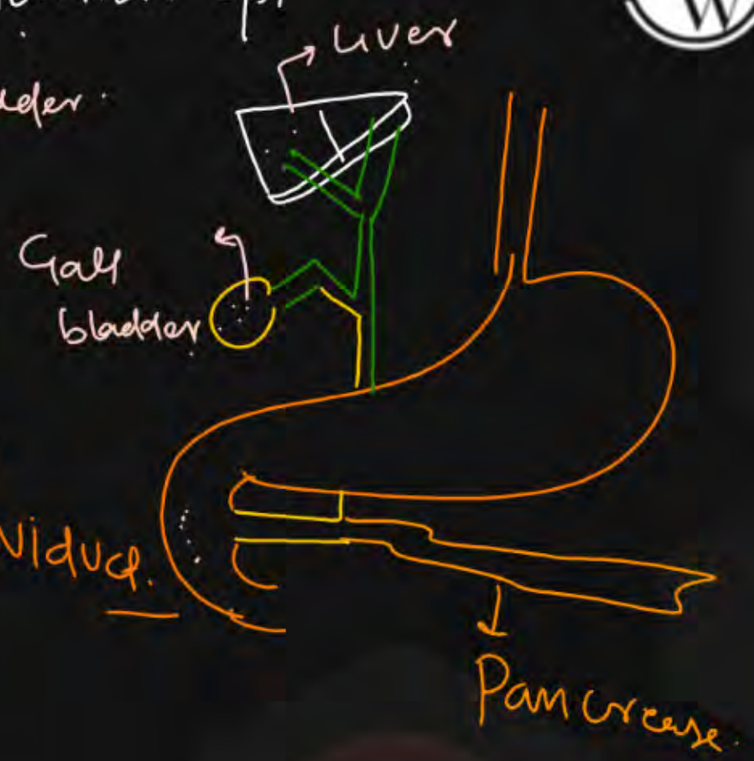
① S. columnar epi + Microvilli = Brushborder S. columnar epi

found in = Gall bladder.

(2) S. Columnar epi + Cilia = Ciliated S. Columnar epi

found in = Falloplan tube | oviduct.



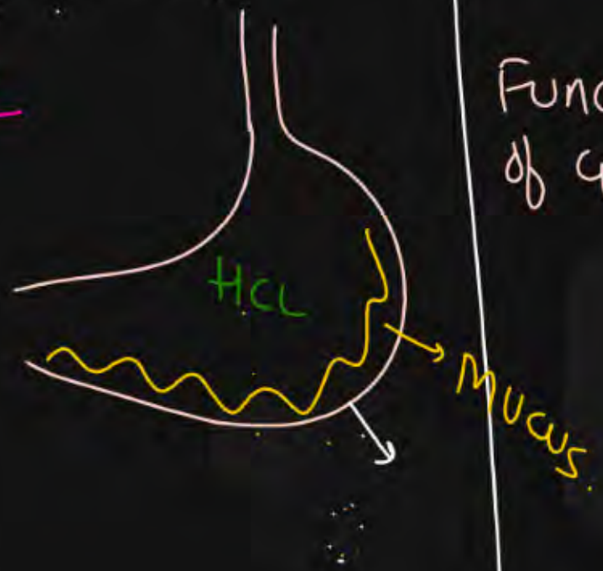


(III) S. columnar epi + Goblet cell = Glandular S. columnar epi

Found in = Stomach

(IV) S. columnar epi + Goblet cell + Microvilli = Brushborder glandular S. columnar epithelium

Found in = Small Intestine



Basic Basis

Goblet cell

Shape of cell = Columnar

Note:- Goblet cell हमेशा Columnar epi में पाया जाता है।

Function of Goblet cell = Secrete Mucus

act as Lubricant

friction से बचाव

Pseudo stratified epithelium [NCERT X]

Pseudo → false; stratified → multilayered

Shape of cell = Columnar.

→ Pseudostratified epi is a type of Simple columnar epi.

↳ Size of cell → ↓ / ↑

Pseudostratified epi + Cilia + Goblet cell = Glandular Ciliated Pseudostratified Columnar epi.

→ Trachea found in: Trachea | Wind pipe

7.1 Animal Tissues
7.2 Organ and Organ System
7.3 Earthworm
7.4 Cockroach
7.5 Frogs

In the preceding chapters you came across a large variety of organisms, both unicellular and multicellular, of the animal kingdom. In unicellular organisms, all functions like digestion, respiration and reproduction are performed by a single cell. In the complex body of multicellular animals the same basic functions are carried out by different groups of cells in a well organised manner. The body of a simple organism like *Hydra* is made of different types of cells and the number of cells in each type can be in thousands. The human body is composed of billions of cells to perform various functions. How do these cells in the body work together? In multicellular animals, a group of similar cells alongwith intercellular substances perform a specific function. Such an organisation is called **tissue**.

You may be surprised to know that all complex animals consist of only four basic types of tissues. These tissues are organised in specific proportion and pattern to form an organ like stomach, lung, heart and kidney. When two or more organs perform a common function by their physical and/or chemical interaction, they together form organ system, e.g., digestive system, respiratory system, etc. Cells, tissues, organs and organ systems split up the work in a way that exhibits division of labour and contribute to the survival of the body as a whole.

7.1 Animal Tissues

The structure of the cells vary according to their function. Therefore, the tissues are different and are broadly classified into four types : (i) Epithelial, (ii) Connective, (iii) Muscular and (iv) Neural.

7.1.1 Epithelial Tissue

We commonly refer to an epithelial tissue as epithelium (pl.: epithelia). This tissue has a free surface, which faces either a body fluid or the outside environment and thus provides a covering or a lining for some part of the body. The cells are compactly packed with little intercellular matrix. There are two types of epithelial tissues namely **simple epithelium** and **compound epithelium**. Simple epithelium is composed of a single layer of cells and functions as a lining for body cavities, ducts, and tubes. The compound epithelium consists of two or more cell layers and has protective function as it does in our skin.

On the basis of structural modification of the cells, simple epithelium is further divided into three types. These are (i) Squamous, (ii) Cuboidal, (iii) Columnar (Figure 7.1).

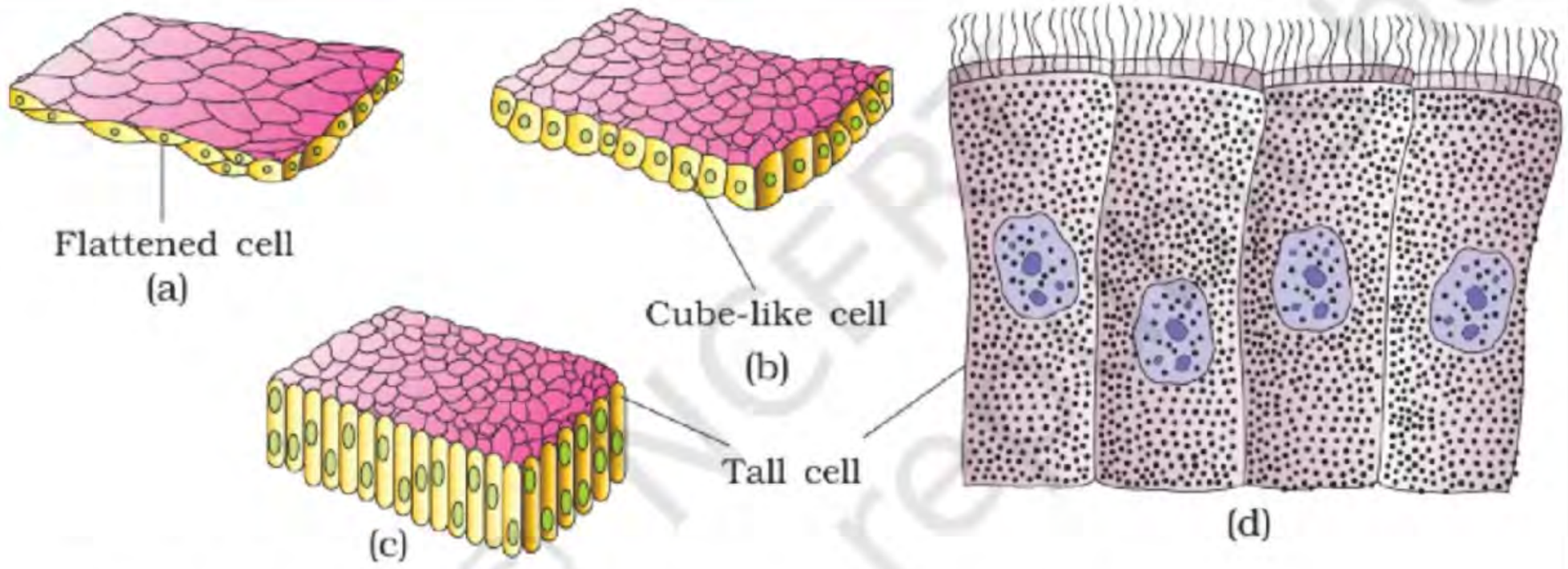


Figure 7.1 Simple epithelium: (a) Squamous (b) Cuboidal (c) Columnar (d) Columnar cells bearing cilia

The **squamous epithelium** is made of a single thin layer of flattened cells with irregular boundaries. They are found in the walls of blood vessels and air sacs of lungs and are involved in functions like forming a diffusion boundary. The **cuboidal epithelium** is composed of a single layer of cube-like cells. This is commonly found in ducts of glands and tubular parts of nephrons in kidneys and its main functions are secretion and absorption. The epithelium of proximal convoluted tubule (PCT) of nephron in the kidney has microvilli. The **columnar epithelium** is composed of a single layer of tall and slender cells. Their nuclei are located at the base. Free surface may have microvilli. They are found in the lining of stomach and intestine and help in secretion and absorption. If the columnar or cuboidal cells bear cilia on their free surface they are called **ciliated epithelium** (Figure 7.1d). Their function is to move particles or mucus in a specific direction over the epithelium. They are mainly present in the inner surface of hollow organs like bronchioles and fallopian tubes.

Blood vessel

Artery vein Capillary

Trachea

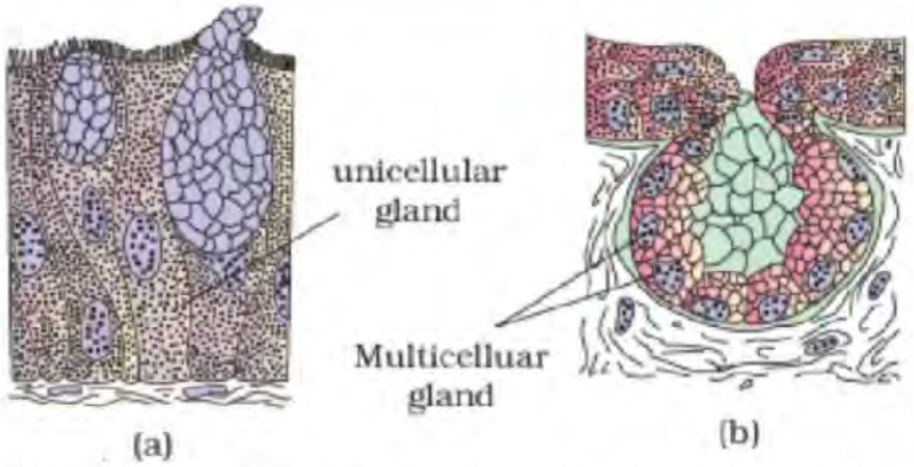


Figure 7.2 Glandular epithelium : (a) Unicellular (b) Multicellular

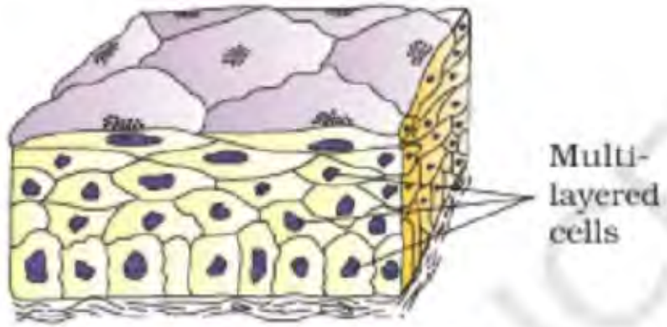


Figure 7.3 Compound epithelium

Gland

Gland का काम ही स्राव i.e Secretion

A cell or group of cell which secrete chemical substance is called gland.

Chemical substance →
- SALIVA
- MUCUS
- Enzyme
- Sweat
- oil | Sebum

⇒ Exocrine gland

- Hormone → Endocrine gland

Some columnar or cuboidal epi get specialised for secretion & hence known glandular epi

Gland

is of two type on the basis number of cell.

Unicellular gland

→ Are made up of single cell.

Expl. Goblet cell, Paneath cell → NCERT X.

Multicellular gland.

→ Are made up multiple cell.

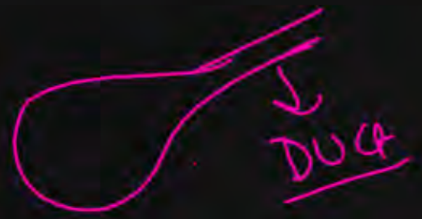
→ Except Goblet & Paneth cell all gland are multicellular gland.

Exp:- Salivary gland
Pituitary gland.

Gland

is of two type on the basis of pouring their secretion

Exocrine gland



→ pour their secretion other than blood.

→ have duct

→ Exp
- S. gland
- Sebaceous gland.
- Sweat gland.
- Digestive gland

Endocrine gland

→ pour their secretion in to the blood.

→ Always secrete hormone.

→ Duct are ~~not~~ absent (Ductless gland.

Exp
- Pituitary gland
- Hypothalamus "
- Pineal "

Some of the columnar or cuboidal cells get specialised for secretion and are called **glandular epithelium** (Figure 7.2). They are mainly of two types: unicellular, consisting of isolated glandular cells (goblet cells of the alimentary canal), and multicellular, consisting of cluster of cells (salivary gland). On the basis of the mode of pouring of their secretions, glands are divided into two categories namely **exocrine** and **endocrine** glands. Exocrine glands secrete mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products. These products are released through ducts or tubes. In contrast, endocrine glands do not have ducts. Their products called hormones are secreted directly into the fluid bathing the gland.

Compound epithelium is made of more than one layer (multi-layered) of cells and thus has a limited role in secretion and absorption (Figure 7.3). Their main function is to provide protection against chemical and mechanical stresses. They cover the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts.

★★★★★

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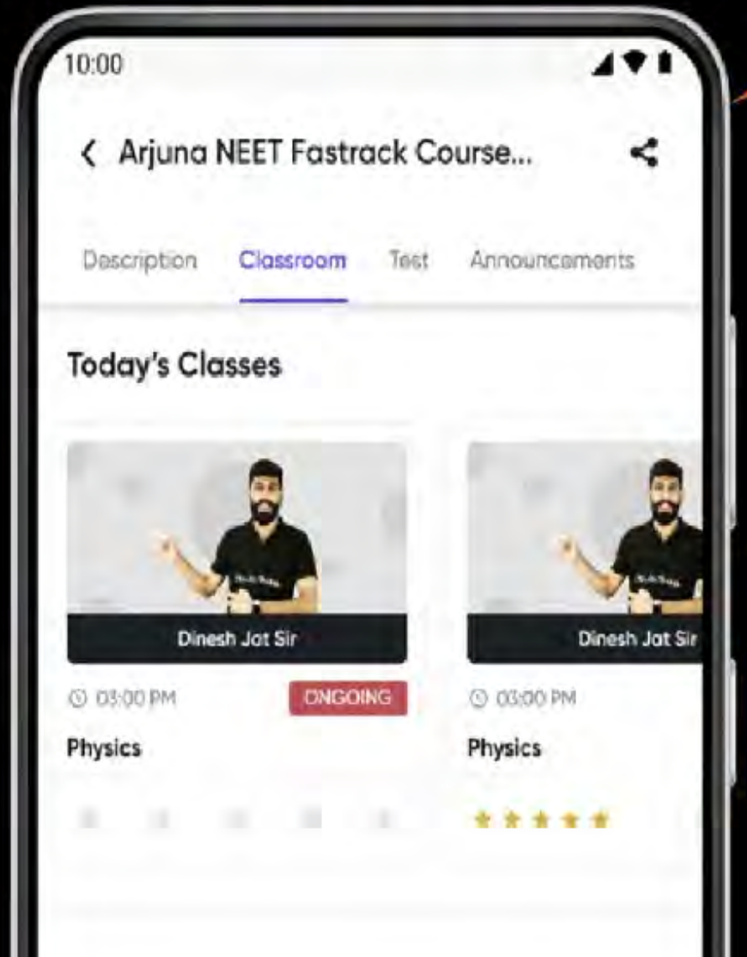

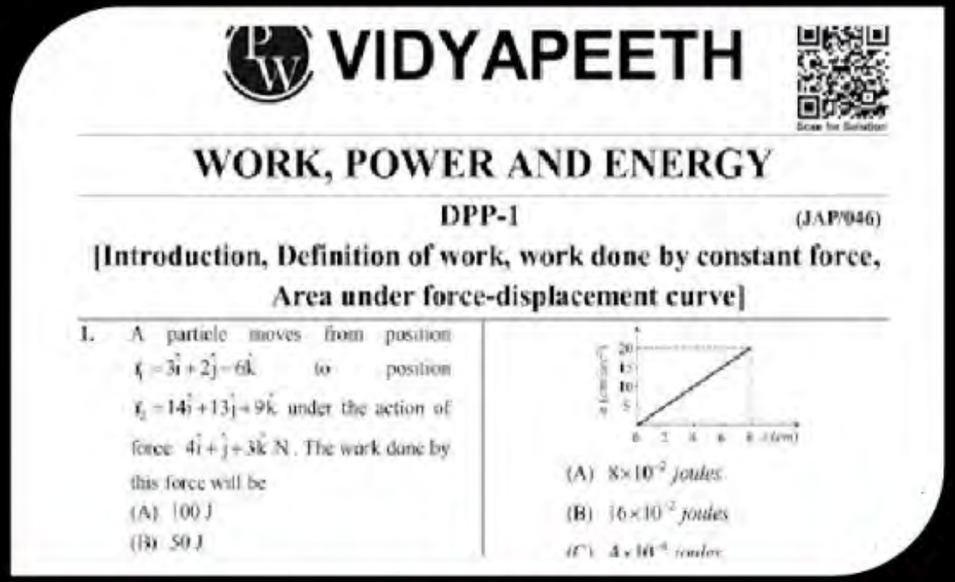

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